# Automated Gravity-Fed Object Counter and Jam Detection System

*STM32F401CCU6 Bare-Metal Prototype for Smart Airport Conveyor Monitoring*

| CONTROLLER | SENSING | TEST RIG | TELEMETRY | RESET INPUT | STATUS OUTPUT |
|---|---|---|---|---|---|
| STM32F401CCU6<br>ARM Cortex-M4 | 2 × TCRT5000<br>PA1 / PA4 analog | 45° chute<br>150 mm gap | HC-05 Bluetooth<br>USART2, 9600 baud | Push button<br>PB5 / EXTI5 | Onboard LED<br>PC13 heartbeat |

## 1. Introduction and Objective

Airport baggage-handling systems require continuous monitoring of object movement, transit speed, dimensions, direction, and line condition. This project develops a compact embedded prototype that counts objects, estimates their speed and physical size, detects reverse movement, and warns an operator about jams or inactivity. A 45° gravity-fed chute is used as a laboratory substitute for a motorized conveyor, with two reflective sensors positioned 150 mm apart along the direction of travel.

## 2. System Architecture

The controller is an STM32F401CCU6 Black Pill programmed in bare-metal C using direct register access without HAL, LL, or CubeMX. Two TCRT5000 reflective IR modules provide analog signals at PA1 and PA4. The documented acquisition path uses ADC1 with DMA2, while a 1 ms SysTick timing engine supports sensor-edge timing, speed calculation, size estimation, and alarm supervision. USART2 connects to an HC-05 Bluetooth module at 9600 baud. A reset pushbutton is connected through EXTI5, the onboard LED provides status indication, and a buzzer provides audible alerts. The sensor modules use the regulated 3.3 V rail and share a common ground with the controller.

## 3. Working Principle

When an object blocks Sensor 1, the controller records the entry time. When the leading edge reaches Sensor 2, speed is calculated from the known separation using v = distance ÷ transit time. The duration for which Sensor 2 remains blocked is then combined with the speed to estimate object length using L = v × blocked time. The measured length is assigned to one of three prototype categories: Small/Carry-on (SM), below 50 mm; Standard (ST), 50–120 mm; or Large/Oversized (LG), above 120 mm. A forward S1-to-S2 passage increments the valid count, while an S2-to-S1 movement is recorded as a direction violation.

## 4. Jam Detection and Operator Feedback

The controller supervises the expected time between the two sensor events. If an object fails to reach Sensor 2 within the permitted transit interval, a jam condition is raised and the buzzer is activated. A separate inactivity timeout identifies a stopped line. Bluetooth telemetry reports the total count, transit count, speed, estimated length, category, alert condition, and violation count to a mobile terminal. Pressing the PB5 reset button clears the counters and tracking state and sends a reset confirmation to the operator.

## 5. Demonstration, Results, and Scope

The documented demonstration sequence checks boot operation, idle inactivity, partial transit, normal forward movement, size classification, jam response, reverse motion, and reset behavior. The logic verification record covers 13 test groups with zero recorded failures. The prototype demonstrates how two analog optical edges can be converted into speed, dimensional categories, direction decisions, and operator alerts. Its scope is limited to a one-object-at-a-time laboratory demonstrator; it does not include motor-speed control, a diverter actuator, persistent storage, or production safety interlocks. Future work can add queue management, encoder or motor control, persistent logging, a dashboard, and validation against known dimensions and a real conveyor system.

## Key Contributions

| MEASUREMENT | SAFETY RESPONSE | OPERATOR VISIBILITY |
|---|---|---|
| Two optical edges produce speed and estimated object length without a camera. | Reverse motion, jams, inactivity, and reset conditions are explicitly handled. | Bluetooth telemetry, LED indication, and buzzer alerts present the line status clearly. |